```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127
Reply from 192.168.2.3: bytes=32 time<1ms TTL=127

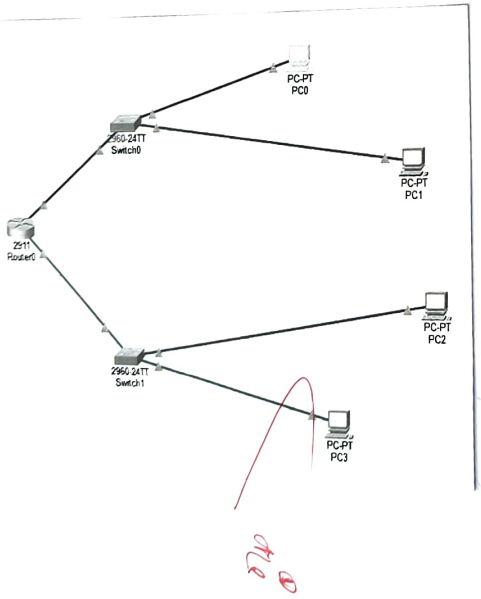
Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.2.1

Pinging 192.168.2.1 with 32 bytes of data:

Reply from 192.168.2.1: bytes=32 time=13ms TTL=255
Reply from 192.168.2.1: bytes=32 time=1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255
Reply from 192.168.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 13ms, Average = 3ms
```



Exc:9

Date: 27/8/26

Implementation of Subnetting in CISCO PACKET TRACER Simulator.

a) Design Multiple Subnets with a suitable Number of Host.

* Aim:

To Design multiple subnets with a suitable number of hosts using subnetting techniques in cisco Packet tracer.

* Steps:

1) Open Cisco Packet Tracer and Create a new project.
2) Design the network topology by planning the required network device
3) Select a suitable network address for the experiment.
4) Calculate the number of subnets required.
5) Determine the no of host needed in each subnet.
6) Calculate the subnet mask using subnetting techniques.
7) Divide the network into multiple subnets.
8) Assign a seperate network address to each subnet
9) Verify the subnet Calculations and host ranges.
10) Save the network topology.

* Result:

Thus, multiple tutorit were successfully assigned according to the required hosts.

b) Assign static IP Addresses across all subnets and connect the subnet through a Router

* Aim:

To assign static IP address to all devices in different subnet and establish communication between the subnet through a router.

* Steps:

1) Connect a router in the switches using appropriate cable.
2) Configure the router interfaces.
3) Assign an IP address and subnet mask to each other router.
4) Configure the switches.
5) Assign static IP address to all computer.
6) Configure the subnet mask for each device.
7) Assign the appropriate default gateway to each computer.
8) Verify the IP configuration of all devices.
9) Save the router configuration.
10) Verify the all devices are properly connected

* Result:

Thus, Static IP addresses were assigned successfully and communication between the subnets was established through the router.

c) Simulate packet Transmission Across the subnets and observe the result:

a) When Subnets are connected through a Router.

* Aim:

To analyse Packet Transmission between different Subnet when a router is used for communication.

1) Open Simulator mode in Cisco packet Tracer.
2) Select a source device from one Subnet.
3) Select a destination device from another Subnet.
4) Generate a packet using the simple PDU tool.
5) Observe the movement of the packet through the network.
6) Analyse how the router receives the packet
7) Observe the forwarding process performed by the router
8) Verify the packet information between different Subnet
9) Use the ping command to test Connectivity.
10) Observe the successful delivery of packets.
11) Records the simulation results.
12) Analyse the Router process.

* Result!

Thus, Packets were transmitted successfully between different Subnets through the router.

b) When Subnets are not Connected through a Router.

* Aim:

To analyse Packet transmission between different subnet when a router is not used for communication.

* Steps:

1) Remove the router from the network topology.
2) Open simulation mode.
3) Select device belonging to different subnets.
4) Generate packets using the simple PDU tool.
5) Observe the packet transmission.
6) Perform a ping test between devices located different subnets.
7) Analyse the Packet flow.
8) Observe the simulator events.
9) Identify the points at which packet transmission fails.
10) Compare the results with the network that uses a router.
11) Record the observation.
12) Analyze the importance of a router in inter network communication.

* Result:

Thus, communication between different subnets failed because a router was not available to forward the packets.